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# A Service-Oriented Real-Time Communication Scheme for AUTOSAR Adaptive Using OPC UA and Time-Sensitive Networking

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**Abstract:** The transportation industry is facing major challenges that come along with innovative trends like autonomous driving. Due to the growing amount of network participants, smart sensors, and mixed-critical data, scalability and interoperability have become key factors of cost-efficient vehicle engineering. One solution to overcome these challenges is the AUTOSAR Adaptive software platform. Its service-oriented communication methodology allows a standardized data exchange that is not bound to a specific middleware protocol. OPC UA is a communication standard that is well-established in modern industrial automation. In addition to its Client–Server communication pattern, the newly released Publish–Subscribe (PubSub) architecture promotes scalability. PubSub is designed to work in conjunction with Time-Sensitive Networking (TSN), a collection of standards that add real-time aspects to standard Ethernet networks. TSN allows services with different requirements to share a single physical network. In this paper, we specify an integration approach of AUTOSAR Adaptive, OPC UA, and TSN. It combines the benefits of these three technologies to provide deterministic high-speed communication. Our main contribution is the architecture for the binding between Adaptive Platform and OPC UA. With a prototypical implementation, we prove that a combination of OPC UA Client–Server and PubSub qualifies as a middleware solution for service-oriented communication in AUTOSAR.

**Keywords:** AUTOSAR adaptive; OPC UA; time-sensitive networking; vehicle communication



**Citation:** Arestova, A.; Martin, M.; Hielscher, K.-S.J.; German, R. A Service-Oriented Real-Time Communication Scheme for AUTOSAR Adaptive Using OPC UA and Time-Sensitive Networking. *Sensors* **2021**, *21*, 2337. https://doi.org/10.3390/s21072337

Academic Editor: Rupak Kharel

Received: 9 March 2021
Accepted: 24 March 2021
Published: 27 March 2021

**Publisher's Note:** MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.



## 1. Introduction

The automotive industry is rushing towards new trends like autonomous driving and Car2X [1–3]. Vehicles are becoming increasingly intelligent and interconnected with the outside world and with each other [4]. The number of new functions in a vehicle is growing rapidly. Particularly in the context of autonomous driving, the need for special sensors that perceive the environment and complex sensor data fusion functions has increased. In addition to the actual control of the vehicle, other components such as driving assistants and infotainment gain in significance. As a consequence, the amount of mixed-critical data to be processed and exchanged has grown enormously.

The current state of the art is that each functionality is implemented on an adapted Electronic Control Unit (ECU). The ECUs are interconnected via proprietary communication buses like CAN and FlexRay [4]. The rapid growth of data and interconnected functions encourage the automotive industry to look for more adaptive and high-end performance software platforms, as well as communication technologies that are able to achieve high throughput, provide determinism, and keep up with the dynamic communication pattern of new features.

The Automotive Open System Architecture (AUTOSAR) (https://www.autosar.org, last accessed on 1 March 2021) is a global development partnership that addresses these